

Robust and Generalizable Decision-Making for CAVs at Unsignalized Intersections

A Deep Reinforcement Learning Approach Reproduction, Benchmarking & Path to
Generalization

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CSE 4610: Design Project

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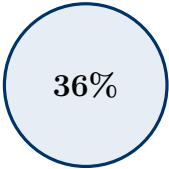
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Outline

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 - ~~• Motivation~~
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The Problem



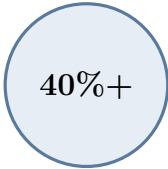
36%

of crashes are
intersection-related [11]



\$371B

US societal cost
of intersection
crashes, 2010 [11]

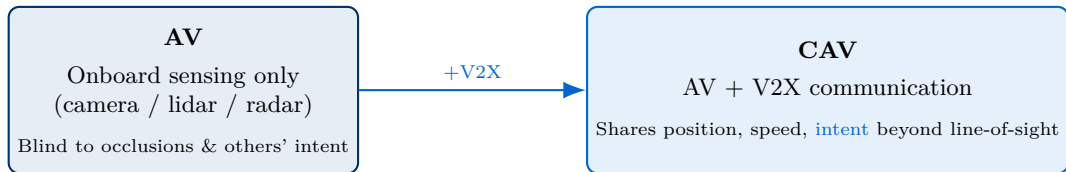


40%+

of EU/US collisions oc-
cur at intersections [2]

- Traffic lights control at the **lane-group** level, not per vehicle
- Fixed timing → stopping even when the road is clear
- Unsignalized crossings: high risk from occlusion & uncertain intent
- No shared protocol → right-of-way must be inferred

Motivation



Signal-Free AIM

Individual, continuous negotiation replaces group-level stopping

Deep RL

Learns adaptive go/yield behavior under stochastic traffic

How should an ego vehicle decide, every timestep, to **go** or **yield** among randomly-behaving vehicles — safely, efficiently, reliably?

- Simultaneous, uncertain decisions among interacting vehicles
- No reservation protocol — right-of-way is inferred, not commanded
- Must generalize across density, driving style, and (eventually) geometry

Key Objectives

Safety

Zero (or near-zero) collisions
across randomized turning
tasks

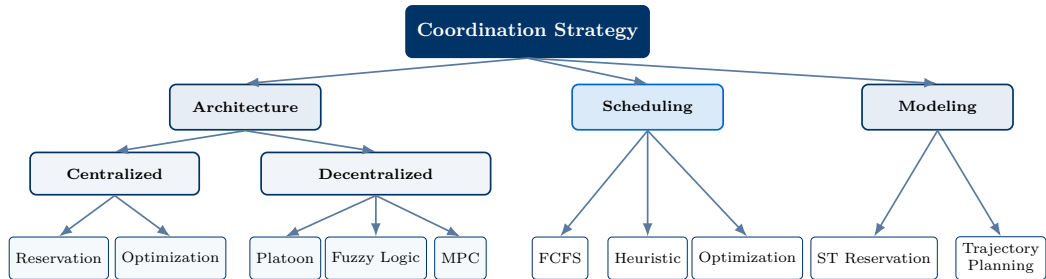
Efficiency

Minimize crossing / travel
time, maximize throughput
speed

Robustness

Stable performance across
traffic densities *and*,
eventually, geometries

Taxonomy of Coordination Strategies



What the Literature Shows

Axis	Most popular	Runner-up
Goal	Safety + Efficiency (37%)	Safety alone (26%)
Architecture	V2V (159 papers)	V2I (119 papers)
Scheduling	Optimization (277 papers)	Heuristic (20 papers)
Modeling	Trajectory Planning	Spatio-Temporal Reservation

RL Literature Thread I: Single-Ego Decision Making

Paper	Idea	Architecture	Key finding
Isele et al. 2018 [4]	DQN vs. TTC + occlusion “creeping”	Single-agent, no comm.	28% faster, <1% collisions
Wu et al. 2019 [9]	Decentralized coordination Q-learning	Multi-agent, sparse comm.	Beats FCFS/LQF/Webster
Kai et al. 2020 [5]	One policy, 3 maneuvers via task vector	Single-agent	Matches/beats 3 separate DQNs
Shu et al. 2022 [8]	Weight transfer across maneuvers	Single-agent	Same-task 100%, cross-task >70%

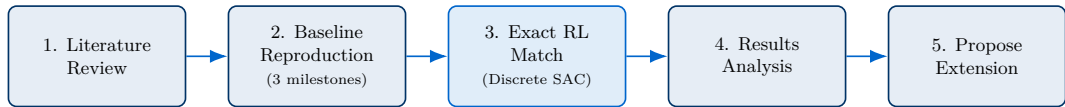
RL Literature Thread II: Graph Representations & Cooperative AIM

Paper	Contribution	Setting
Chen et al. 2025 [1]	Scenario-agnostic “lane-gap” + PPO + CILQR	Single-ego, multi-scenario (93–97%)
Schier et al. 2023 [7] Klimke et al. 2022 [6]	Vehicle+road graph, learned edges Edge-augmented RGCN + TD3, controls <i>all</i> vehicles	Predicts right-of-way at 99.5% acc. True cooperative AIM
Huamán Reyna et al. 2026 [3]	Survey, 53 papers	Hybrid methods score best overall

Where we fit

- Single-ego decision-making, multi-task state (like Kai et al., Xiao et al.)
- Not centralized AIM (Klimke et al.) — tractable scope, still comparable

Methodology at a Glance



Each stage builds on the previous milestone's validated pipeline — detailed next.

Reproduction Target: Xiao et al. (2024, IEEE T-VT)

“Decision-Making for AVs in Random Task Scenarios at Unsignalized Intersection Using Deep RL” [10]

Three contributions

1. **Mix-Attention Network** (replaces plain MLP)
2. **Task-intent variable** μ_a
3. **Segmented replay buffer**

Setting

4-way, single-lane, unsignalized; random task/episode; **highway-env**; Discrete SAC, 50,000 episodes

Disclaimer: Scope of This Work

Reproduction & benchmarking study — not our primary contribution

Build a working pipeline, understand the environment deeply, then propose a novel direction.

Acknowledged limitations

- Original source code **not released**
- Some hyperparameters / reward constants only partially specified
- Reproduction is faithful, but **not byte-for-byte**

Research Progression: Three Milestones

M1 — Basic Ring Stabilization

Single-agent PPO, 230m ring, 21 IDM cars

Result: damped phantom shockwaves



M2 — Phase 1 Exploratory Reproduction

PPO, 4-way intersection, Original vs. Config B ($+\mu_a$)

Result: found left-turn vulnerability; fixed eval bug



M3 — Phase 2 Exact Reproduction

Discrete SAC (twin critics, auto-tuned α), 50k episodes

Result: 100% success, 0% collisions (1,050 tests)

State Space Formulation

Observation vector (48 m radar radius, up to 9 nearest surrounding vehicles):

$$\mathbf{S} = [S_{\text{ego}}, S_1, S_2, \dots, S_9]$$

Ego Vehicle State ($S_{\text{ego}} \in \mathbb{R}^6 \text{ or } 7$)

$$\mathbf{S}_{\text{ego}} = [p_{\text{ego}}, x, y, v_x, v_y, \phi, (\mu_a)]$$

- $p_{\text{ego}} \in \{0, 1\}$: **Existence flag** (1 = present)
- x, y : **Global position** coordinates (m)
- v_x, v_y : **Absolute velocity** along x, y (m/s)
- ϕ : **Heading angle** / orientation (rad)
- μ_a : **Task-intent feature** (*Config B only*)
Encodes route (left/straight/right) & exit progress

9 Surrounding Vehicles ($S_i \in \mathbb{R}^6$ each)

$$\mathbf{S}_i = [p_i, \Delta x_i, \Delta y_i, \Delta v_{x,i}, \Delta v_{y,i}, \phi_i]$$

- $p_i \in \{0, 1\}$: **Presence flag** (0 if empty / padded)
- $\Delta x_i, \Delta y_i$: **Relative position** ($x_i - x, y_i - y$) (m)
- $\Delta v_{x,i}, \Delta v_{y,i}$: **Relative velocity**
($v_{x,i} - v_x, v_{y,i} - v_y$) (m/s)
- ϕ_i : **Heading angle** of vehicle i (rad)
- **Ordering**: Sorted ascending by distance
 $d_i \leq 48$ m

Total Dimensionality & Intent Role

Original (Baseline): $6 + (9 \times 6) = \mathbf{60}$ -dim

Config B (Proposed): $7 + (9 \times 6) = \mathbf{61}$ -dim (+ μ_a)

Why μ_a matters: Conditions speed on the upcoming maneuver (straight/left/right) and exit progress, enabling adaptive conflict negotiation without separate per-task networks.

Action Space & Reward Function

Action (discrete, 3 values)

$$A = \{\text{accel}, \text{idle}, \text{decel}\}$$

$$a = K_{P,A}(v_{tar} - v), \quad K_{P,A} = 5$$

Lateral control is a fixed PD controller

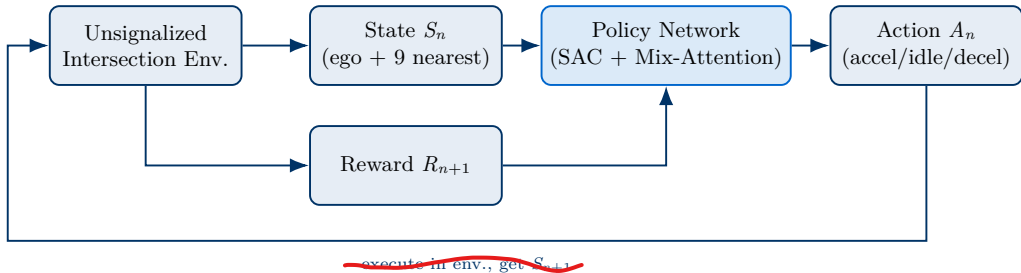
— RL controls only longitudinal speed.

Reward

$$R = \begin{cases} +30 & \text{reach destination} \\ -120 & \text{collision} \\ +0.1 & \text{reach target speed} \end{cases}$$

- Sparse terminal terms (safety + completion)
- Small dense term rewards cruising
- Collision penalty $\gg$ arrival reward

System Pipeline



Discrete Soft Actor-Critic (Discrete SAC)

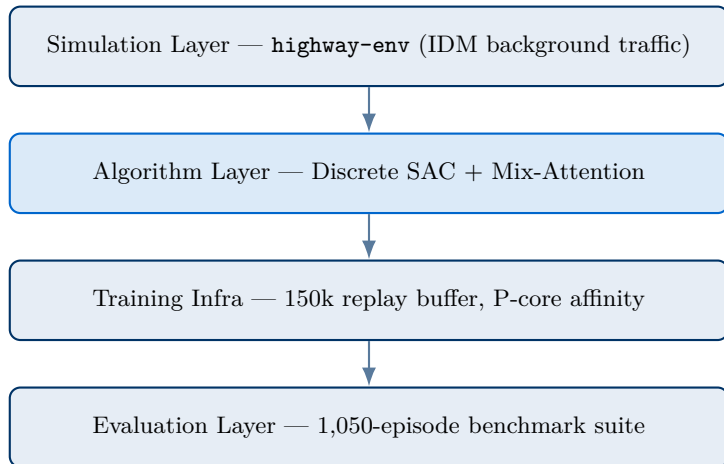
Why SAC over PPO

- Off-policy $\rightarrow$ sample efficient
- Max-entropy $\rightarrow$ broad exploration
- Auto-tuned α — no fragile hyperparameter

$$\pi^* = \arg \max_{\pi} \sum_n \mathbb{E}[\gamma^n (R_n + \alpha \mathcal{H}(\pi(\cdot|S_n)))]$$

- Twin Q-critics reduce overestimation bias
- Discrete policy: softmax over 3 actions

Implementation Stack

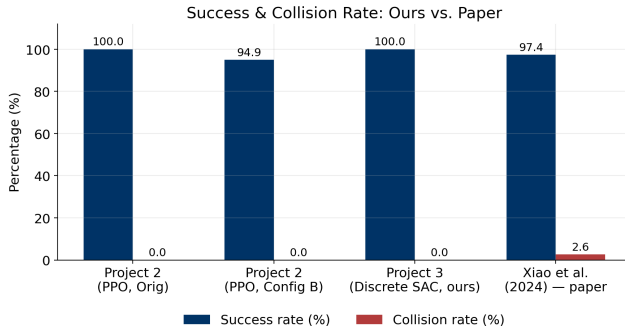


Simulation & Training Setup

	Project 1: Ring	Project 2: Phase 1	Project 3: Phase 2	Paper
Scenario	230m ring	4-way	4-way	4-way
Sim. engine	Flow + SUMO	Flow + SUMO	highway-env	highway-env
Algorithm	PPO	PPO	Discrete SAC	Discrete SAC
Network	Actor-Critic MLP	[32, 32, 32]	[256, 256, 64, 3]	[256, 256, 64, 3]
Obs. space	3-dim POMDP	60 / 61-dim	60 / 61-dim	60 / 61-dim
Action space	Continuous [−1, 1] m/s ²	Discrete(3)	Discrete(3)	Discrete(3)
Training budget	123 iters	2.5k / 9.36k iters	50k episodes	50k episodes

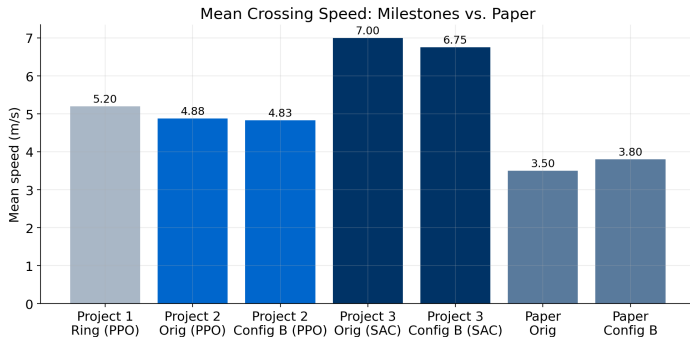
~~$\gamma = 0.90$, $\tau = 0.005$, replay = 150,000, batch = 1024 — matched for Project 3.~~

Result 1: Success & Collision Rate



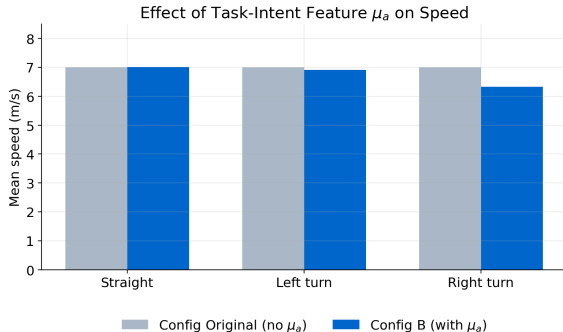
100% success, 0% collisions over 1,050 episodes — exceeds the paper's 97.4%/2.6%.

Result 2: Mean Crossing Speed



Nearly **2× faster** crossing than the paper — the simulators are not physically equivalent.

Result 3: Intent-Driven Speed Modulation



Config B ($+\mu_a$) slows for turns (Left 6.91, Right 6.32 m/s); Original stays uniform (~ 7.00 m/s).

The Simulation Physics Gap: 100% vs. 97.4% Success

1. Micro-yielding

- IDM vehicles yield to an assertive AV
- Paper's rigid model doesn't soften

2. Exposure window

- Ours: ~ 3.5 s in the conflict box
- Paper: ~ 10 s at a $\sim 2\text{--}3.8$ m/s crawl
- Shorter exposure $\rightarrow$ fewer collisions

Takeaway

Simulator physics difference — not an algorithmic bug.

Challenges and Limitations

Incomplete Spec

Paper's code & some hyperparameters unavailable

Multi-Ego Eval Bug

Caught & fixed during Milestone 2

Partial Coverage

Only 2/8 paper configs matched so far

Hardware Bottleneck

Hybrid-CPU thread stalls — fixed via P-core affinity

Sim-to-Sim Gap

highway-env vs. paper's physics differ

Single Geometry

Only one 4-way layout tested to date

Proposed Novelty: Geometry-Agnostic Decision-Making

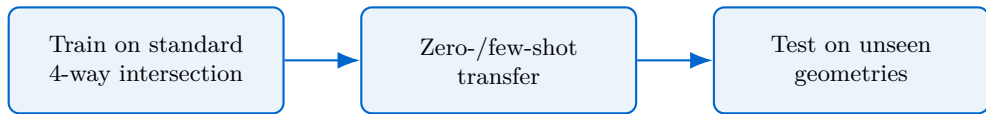
Core direction

Every model above is trained/evaluated on *one* fixed 4-way geometry — real deployments include T-junctions, merges, roundabouts.

Goal: a policy that does not depend heavily on its training geometry.

- Evaluate: 4-way, T-junction, ramp merge, roundabout
- Measure transfer of right-of-way reasoning, not just re-training

Generalization Strategy



~~Unseen geometries: T-junction · ramp merge · roundabout~~

~~Mirrors Klinke et al. [6]: edge-aware graph representations generalize far better across intersection sizes than vertex-only models.~~

Immediate Next Steps

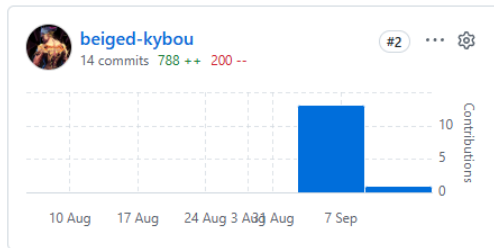
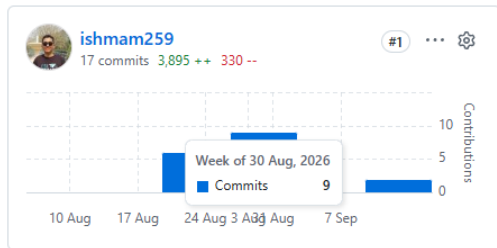
1. Finish remaining 6/8 configurations
2. Build T-junction & roundabout scenes
3. Zero-shot evaluate Project-3 policy on unseen geometries
4. Explore graph / relational state encoding
5. Quantify the sim-to-sim physics gap

Conclusion

- Full DRL pipeline validated across three milestones
- 1:1 Discrete SAC reproduction [10], matching/better outcomes
- Result gap traced to simulator physics, not a bug
- Concrete path proposed toward [geometry-agnostic](#) decision-making

Thank you — Questions?

GitHub Contributors Activity Graph



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